

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|--------------------------------|--|
| 1. Vertex _____ | (A) The U-shaped graph of a quadratic function. |
| 2. Axis of symmetry _____ | (B) The expression $b^2 - 4ac$; its sign tells the number and type of solutions. |
| 3. Zero (root) _____ | (C) The turning point of a parabola, where it reaches its maximum or minimum. |
| 4. Discriminant _____ | (D) The number $i = \sqrt{-1}$, which satisfies $i^2 = -1$. |
| 5. Imaginary unit _____ | (E) An x -value where $f(x) = 0$; an x -intercept of the parabola. |
| 6. Completing the square _____ | (F) The vertical line $x = h$ that splits a parabola into mirror halves. |
| 7. Parabola _____ | (G) A quadratic written as $ax^2 + bx + c$. |
| 8. Standard form _____ | (H) Rewriting $x^2 + bx$ as a perfect-square trinomial by adding $(\frac{b}{2})^2$. |

Part B — Multiple Choice (12 pts)

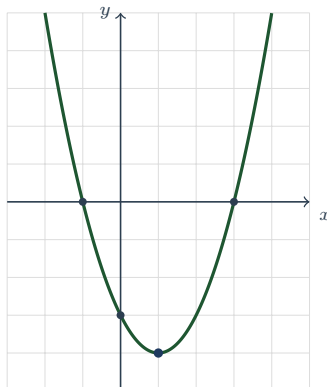
Circle the best answer.

- The vertex of $y = (x - 4)^2 - 1$ is (A) $(4, -1)$ (B) $(-4, -1)$ (C) $(4, 1)$ (D) $(-4, 1)$
- $\sqrt{-25} =$ (A) $5i$ (B) -5 (C) $25i$ (D) 5
- The discriminant of $x^2 - 4x + 7 = 0$ is -12 , so the equation has (A) two real solutions (B) one real solution (C) two complex (non-real) solutions (D) no solutions of any kind
- Factored completely, $x^2 - 49$ is (A) $(x - 7)(x + 7)$ (B) $(x - 7)^2$ (C) $(x + 7)^2$ (D) prime
- The axis of symmetry of $y = x^2 - 6x + 5$ is (A) $x = 3$ (B) $x = -3$ (C) $x = 6$ (D) $x = 5$
- A line and a parabola can intersect in *at most* how many points? (A) 1 (B) 2 (C) 3 (D) infinitely many

Part C — Short Answer & Computation (40 pts)

Show your work.

- Read the graph.** The parabola f is shown below.



Vertex: _____; axis of symmetry: _____; y -intercept: _____;
 zeros: _____; minimum *or* maximum value: _____; range: _____

2. **From standard form.** For $f(x) = x^2 - 6x + 5$, use $x = -\frac{b}{2a}$ to find the vertex, give the axis of symmetry, and state whether the parabola opens up or down.

3. **Solve by factoring.**

$$x^2 + 2x - 15 = 0$$

4. **Solve using square roots.**

$$(x - 2)^2 = 49$$

5. **Solve by completing the square.**

$$x^2 + 6x - 7 = 0$$

6. **Complex numbers.** Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$. Write each answer in $a + bi$ form.

(a) $z_1 + z_2 =$ _____ (b) $z_1 - z_2 =$ _____

(c) $z_1 \cdot z_2 =$ _____

7. **Quadratic formula & discriminant.** Solve $2x^2 + 3x - 2 = 0$. First compute the discriminant and state how many real solutions to expect, then solve.

8. **System.** Solve the linear-quadratic system algebraically. Give every solution as an ordered pair.

$$y = x^2 - 2x - 3 \quad y = x - 3$$

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **Projectile.** A ball is thrown upward from a rooftop. Its height (in feet) after t seconds is

$$h(t) = -16t^2 + 32t + 48.$$

(a) From which feature of the model do you read the ball's *starting height*? Give that height. (b) Find the *maximum* height and the time it occurs. Which feature of the graph is this? (c) When does the ball hit the ground? Which feature answers this, and why do you reject the other solution?

2. **Discriminant reasoning.** Consider the equation $x^2 - 4x + k = 0$, where k is a constant. (a) Write the discriminant in terms of k . (b) For what values of k does the equation have two real solutions? one? two complex solutions? (c) Explain how the discriminant is connected to the number of x -intercepts of the parabola $y = x^2 - 4x + k$.